

PHASE 4

Project Planning Phase

AI StudyBuddy – AI-Powered Learning and Study Support Platform

Project Focus
AI-assisted learning resources, secure backend services, and personalized study support.

Phase Objective

To organize the implementation work into manageable activities, define the development environment, and establish the configuration needed before coding.

4.1 Project Setup Plan

- Create the main project directory as AI StudyBuddy.
- Open the project in Visual Studio Code.
- Initialize the Node.js project.
- Install the backend and AI-related packages.
- Add a development-time utility such as Nodemon.
- Prepare the source files and folders for modular development.

4.2 Planned Dependencies

- express – backend web framework.
- mongoose – MongoDB object modeling and validation.
- bcryptjs – password hashing.
- jsonwebtoken – token-based authentication.
- cors – cross-origin request support.
- dotenv – environment configuration.
- multer – handling uploaded files.
- @google/genai – communication with the Gemini service.
- nodemon – convenient development server restarts.

4.3 Initial File Plan

- index.js as the application entry point.
- .env for environment-specific configuration.
- package.json for project metadata and dependencies.
- .gitignore to exclude files that should not be committed.

4.4 Configuration Plan

- PORT is used to define the server port.
- MONGO_URI stores the MongoDB connection string.
- JWT_SECRET provides the signing secret for authentication tokens.
- GEMINI_API_KEY stores the credential required for Gemini access.

4.5 Planned Development Areas

- Application startup and database connectivity.
- Authentication and authorization middleware.
- Study-material upload and management.
- AI service integration.

- Database schemas and persistence.
- REST API routes.
- Administrative endpoints.
- API verification and execution demonstration.

4.6 Planning Result

The planning stage establishes a modular implementation path in which security, data storage, AI processing, API routing, and administration can be developed and checked as separate but connected parts.